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33-151
Project Proposal
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Modeling MOFs and Carbon Capture through Potential Energy Diagrams:
"Concepts of a Plan: Saving Climate Crisis"

In this project, we're interested in exploring the molecular characteristics of Metal Organic Frameworks (MOFs) and the implications they have for the climate crisis. MOFs are porous crystalline materials with a high surface area and a 3-dimensional cage-like grid, making them a capable candidate for carbon capture. They enable mixed gases to diffuse with the pore environment of an MOF specifically targeting one molecule; research has already suggested they could capably engage in adsorption of greenhouse gases such as CO2 and CH4, and by modeling the potential energy of the interactions between these MOFs and greenhouse gases, we aim to gain a better understanding of the viability of MOFs as a way to reduce levels of greenhouse gases in Earth's atmosphere. For the excitement of this project, we would like to model the speedy elimination of either CO2 or CH4 from the atmosphere in the span of one day, partly curing the world of climate change, and saving life on Earth. 🙂 *#cureclimatechangeinoneday*

This project aims to simplify the potential energy modeling to a macro-level analysis; there's no intention of considering the internal interactions of the MOFs in our system due to their complexity. Further research will determine which interactions are relevant for our use. The principles of the Density Functional Theory will be used to integrate spatial electron density function (obtained from preexisting data) with respect to R, over a volume– a concept similar to that of Wednesday's lecture– and use those calculations to find the electrostatic force at each point along the surface of interaction between the particles. A model and plot of the potential energy can then be produced relative to the spatial arrangement of the particles. This project uses theoretical physics in releasing MOFs into the atmosphere for carbon capture at a macro scale, along with computation in potential energy calculations *ab initio* and subsequent PE modeling with vPython applications. This project will be challenging, but also super engaging and exciting to work with!